

User Manual: AI-Based Early Disease Risk Prediction System

Welcome to the **AI-Based Early Disease Risk Prediction System**. This system utilizes Machine Learning to predict the risk of early-onset Diabetes and Cardiovascular Disease based on user-provided health metrics.

This manual provides a comprehensive, step-by-step guide to installing the required software, setting up the project environment, and running the application. It is designed so that anyone can follow along without prior experience.

1. Prerequisites

Before starting, ensure you have the following installed on your system:

1.1 Python Installation

The project requires Python 3.9 or higher.

- Download Python:** Go to the official website [python.org/downloads](https://www.python.org/downloads/) (<https://www.python.org/downloads/>) and download the latest stable release for your operating system (Windows/macOS/Linux).
- Installation (Windows):** Run the installer. **CRITICALLY IMPORTANT:** Check the box that says "Add Python to PATH" at the bottom of the installation window before clicking "Install Now".
- Verify Installation:** Open your terminal or command prompt and type:

```
python --version
```

It should return the version installed (e.g., Python 3.10.x).

1.2 Integrated Development Environment (IDE)

An IDE is a software application that provides comprehensive facilities to computer programmers for software development. We recommend **Visual Studio Code (VS Code)** for this project.

- Download VS Code:** Go to code.visualstudio.com (<https://code.visualstudio.com/>) and download the installer for your OS.
 - Install:** Run the installer and follow the standard installation prompts.
 - Recommended Extensions for VS Code:**
 - Python (by Microsoft)
 - Jupyter (by Microsoft) - Required if you wish to run the model training notebooks.
-

2. Project Setup

2.1 Get the Project Folder

Ensure you have the project folder named FYRP (or Final-year-Research-Project) on your local machine. You can place this folder anywhere (e.g., your Desktop or Documents).

2.2 Open the Project in VS Code

1. Open **Visual Studio Code**.
2. Go to **File > Open Folder...** (or `Ctrl+K Ctrl+O`).
3. Select the `FYRP` folder.

2.3 Set Up a Virtual Environment (Recommended)

A virtual environment isolates this project's libraries from other Python projects on your system.

1. Open a new terminal in VS Code by going to **Terminal > New Terminal** from the top menu.
2. Create the virtual environment by running the following command:

```
python -m venv .venv
```

3. Activate the virtual environment:

- o **Windows:**

```
.venv\Scripts\activate
```

- o **macOS / Linux:**

```
source .venv/bin/activate
```

(You should see `(.venv)` appear at the beginning of your terminal prompt, indicating it is active).

3. Installing Dependencies (Libraries)

The project relies on several third-party Python libraries for data processing, machine learning, and the web interface.

With your virtual environment activated, install the required packages using the terminal. You can install them all at once using the following command:

```
pip install streamlit pandas scikit-learn xgboost optuna imbalanced-learn seaborn matplotlib joblib tqdm
```

Breakdown of Libraries Used:

- **streamlit**: Used to build the interactive web dashboard.
- **pandas**: Used for data manipulation and analysis.
- **scikit-learn**: Core machine learning library used for modeling and preprocessing.
- **xgboost**: Advanced gradient boosting algorithm used for high-performance models.
- **optuna**: Used for hyperparameter tuning in model notebooks.
- **imbalanced-learn (imblearn)**: Provides SMOTE for handling imbalanced datasets.
- **seaborn & matplotlib**: Used for data visualization and graphing.
- **joblib**: Used for saving and loading the trained machine learning models (`.pkl` files).
- **tqdm**: Displays progress bars for longer-running tasks.

(Note: Depending on your internet speed, downloading and installing these libraries may take a few minutes).

4. Running the Web Application

The front-end user interface is built with Streamlit. To start the application and use the prediction system:

1. Ensure your terminal is open in the root folder of the project (FYRP) and the virtual environment is activated (`.venv`).
2. Run the following command to start the Streamlit server:

```
streamlit run app/app.py
```

3. The terminal will display a local network URL, and your default web browser will automatically open a new tab pointing to:

```
http://localhost:8501
```

4. Using the App:

- Use the sidebar navigation on the left to switch between "Home", "Diabetes Prediction", "Cardiovascular Prediction", and "Model Information".
- In the prediction pages, fill out the medical parameters in the form and click **Predict Risk** to receive your AI-driven risk assessment and health recommendations.

5. Working with Machine Learning Models (Optional)

If you are a developer or researcher looking to view how the models were trained, evaluated, and saved, you can explore the Jupyter Notebooks.

1. Ensure you have the **Jupyter** extension installed in VS Code.
2. Navigate to the `notebooks/` folder in the VS Code file explorer.
3. Open either:
 - `diabetes_model.ipynb`
 - `cardiovascular_model.ipynb`
4. You can run the code cells sequentially to see the data processing, model training, hyperparameter tuning, and export of the `.pkl` files to the `models/` folder.

6. Project Structure Overview

Here is a brief overview of the project structure you downloaded, so you know where everything is located:

```
FYRP/
├── .venv/                # Virtual environment folder (created by you)
├── app/
│   └── app.py            # The main Streamlit web application file
├── data/
│   ├── cardiovascular.csv # Dataset for Cardiovascular disease
│   └── diabetes.csv       # Dataset for Diabetes
├── models/
│   ├── cardiovascular_model.pkl # Trained ML model artifact
│   └── diabetes_model.pkl       # Trained ML model artifact
└── notebooks/
    ├── cardiovascular_model.ipynb # Training script/notebook
    └── diabetes_model.ipynb       # Training script/notebook
```

(Note: Any internal configurations or folders not listed above can be ignored for standard usage).

Troubleshooting

- **"Command not found: python"**: Ensure you added Python to your system's PATH during installation. Try using `python3` instead of `python`.
- **"ModuleNotFoundError"**: Ensure your virtual environment is activated and you ran the `pip install` command successfully for all required libraries.
- **Port 8501 is already in use**: If Streamlit cannot start on port 8501, it will automatically try 8502. Just follow the link provided in your terminal output.